Assignment

Module 3.4 Network Cable Types and Tools Theory

1. **List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).**

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Feature	Ethernet Cable	Fiber Optic Cable
1. Data transmission	Transfers data using electrical signals through copper wires.	Transmits data using light signals through glass or plastic fibers.
2. Speed and distance	Usually used for shorter connections, with standard Ethernet cables supporting distances of up to 100 meters.	Supports high-speed data transfer over much longer distances, sometimes spanning several kilometers.
3. Cost and interference	Generally, it is less expensive and easier to install but can be affected by electromagnetic interference.	Usually more expensive to install, but it is not affected by electromagnetic interference.

Real-World Examples

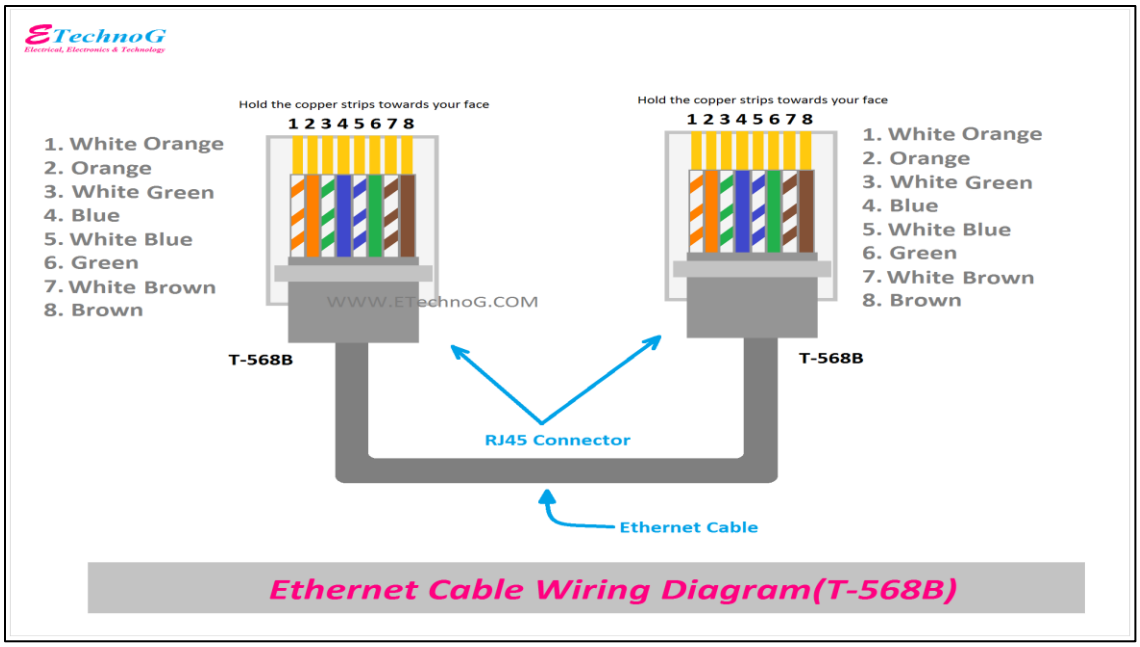
1. Ethernet cable – Home network: An Ethernet cable links a Wi-Fi router to a computer or smart TV, giving us a steady and dependable internet connection.

2. Fiber optic cable – Data center: Fiber cables link servers and network switches, letting huge amounts of data move fast between devices.

3. Fiber Optic cable – College campus: Fiber cables link different campus buildings, giving fast and dependable network access across long distances.

2. **Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.**

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3. **Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.**

➢ **Materials Required**

CAT6 Ethernet cable

CAT6-compatible RJ45 connector

RJ45 crimping tool

Wire stripper and cable cutter

Ethernet cable testing device

Procedure

1. Cut the cable: Measure the desired cable length and use a cutter to cut the CAT6 cable cleanly.

2. Remove the outer jacket: Strip about 2–3 cm of the cable's outer insulation, taking care not to damage the inner wires.

3. Separate the wire pairs: Gently untwist the four pairs and straighten all eight wires.

4. Follow the T568B color code: Arrange the wires in the correct order as shown below.

Pin number	Wire color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

5. Trim the wires: Keep all eight wires in the correct sequence and cut their ends evenly so they fit inside the connector.

6. Insert the wires into the RJ45 plug: Carefully push each wire into its designated channel until the wires reach the front of the plug. Ensure that part of the outer jacket enters the connector to provide proper support.

7. Crimp the connector: Position the RJ45 plug in the crimping tool and press the handles firmly so the metal contacts connect with the wires and secure the plug.

8. Inspect the plug: Check that all wires are correctly positioned, fully inserted, and firmly held in place.

9. Test the cable: Use an Ethernet cable tester to verify that all eight pins are connected correctly and that there are no wiring faults.

10. Finish the other end: For a straight-through Ethernet cable, install another RJ45 connector at the opposite end using the same T568B wiring arrangement.

4. **Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.**

- For a gaming café with 10 computers, I would pick Ethernet cables (CAT6) because they give fast and steady connections for LAN gaming without costing too much. The cables stay short enough for Ethernet to work well between the computers and the network switch. I would use an RJ45 crimping tool to put connectors on the cables. Fiber optics work better for long distances, but Ethernet is the cheap choice for a small gaming café.